Niranjan Sai KV

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Education

University of Texas at Dallas **Richardson, Texas** *01/2024 – 05/2025*
MS, Information Technology and Management

Indian Institute of Space Science and Technology **Trivandrum, India** *08/2015 – 05/2019*
B.Tech, Electronics and Communication Engineering

Skills

- **High Performance Computing (HPC):** Linux (RHEL/Rocky, Ubuntu), Cluster Management (OpenHPC, BaseCommand, Confluent), Job Scheduling (Slurm, PBS, SGE), Parallel File Systems (Panasas, GPFS, WekaFS, LustreFS, MooseFS)
- **Automation & Observability:** Go, Python, Bash, Ansible, Git, Splunk, Zabbix, Prometheus, Grafana, PowerBI
- **Platform Engineering:** Docker, Apptainer, Kubernetes, LXC/LXD, Proxmox, VMware, Citrix, KVM
- **Systems, Security & Operations:** Cloud Computing, Networking, Vulnerability Assessment, Compliance Audits, IT Project Management, Agile Methodology, Database Management

Certifications:-
Redhat Certified System Administrator, Comptia Cybersecurity Analyst+, AWS Cloud Solutions Architect

Professional Experience

Linux (HPC) Engineer *X-ISS (N2 Group)* **Richardson, Texas** *07/2025 – Present*
- Designed, deployed, and operated HPC platforms integrating multiple schedulers, scientific applications, and cluster management tools; troubleshot complex system issues and optimized infrastructure to maintain high availability, scalability, and performance.
- Built a low-overhead Go-based node agent for on-premise and cloud HPC clusters to stream real-time telemetry (CPU, GPU, storage, network), powering a centralized SaaS observability platform for monitoring, alerting, and data-driven capacity planning.
- Optimized HPC cluster performance using LINPACK, IOR, nvprof, and strace to benchmark and profile compute, GPU, and storage subsystems, driving improvements in throughput and resource utilization.
- Delivered HPC application environments via modules, containers, and Spack, while enabling user-friendly access through Open OnDemand, DCV, and VNC, improving usability and researcher productivity.

HPC Engineer Intern, *University of Texas at Dallas* **Richardson, Texas** *06/2024 – 05/2025*
- Implemented end-to-end HPC observability by developing a Python-based head node agent and integrating Node, Redfish, Slurm, and DCGM exporters, enabling system-wide monitoring, Grafana dashboards, and automated alerting via Prometheus.
- Engineered Ansible playbooks to automate end-to-end HPC cluster provisioning using OpenHPC and Slurm, integrating Open OnDemand for interactive user access and XDMoD for workload performance analytics and reporting.
- Built RPM packages for scientific software and maintained a local repository for HPC clusters, enabling consistent deployment through environment modules and Apptainer container definitions to ensure reproducible research workflows.

IT Infrastructure Engineer *Indian Space Research Organisation* **Sriharikota, India** *08/2019 – 12/2023*
- Built and managed a VMware vSphere-based private cloud platform, migrating 150+ mission-critical applications and driving 65% improvement in resource utilization through capacity engineering and centralized SAN/NAS storage architecture.
- Architected a self-hosted GitLab platform and automation framework to track configuration changes across infrastructure, complemented by NetVault-based archiving for secure backup, retention, and disaster recovery.
- Implemented hyper-converged infrastructure and network designs for CCTV recording and streaming workloads, enabling scalable, high-throughput video processing and optimized storage performance.
- Engineered an internal collaboration platform using open-source tools (Zimbra, Nextcloud, Jitsi), delivering integrated services for email, file sharing, and video conferencing across the organization.
- Built and operated secure, fault-tolerant data pipelines for streaming real-time telemetry from rocket integration and launch facilities, including meteorological instrumentation, ensuring reliable ingestion, processing, and monitoring of mission-critical data.
- Deployed and integrated enterprise observability platforms (Splunk, Zabbix, Nagios, Graylog) across 200+ hosts, implementing alert correlation and noise reduction to improve signal quality and reduce incident response time by 40%.
- Managed enterprise security operations spanning EDR and firewall controls, and defined incident response playbooks aligned with MITRE ATTCK and CIS benchmarks to enhance compliance posture, audit readiness, and risk mitigation.
- Administered and upgraded a high-performance computing (HPC) cluster using Slurm, Confluent Cluster Manager, and Lustre, supporting workloads for weather prediction and Ansys-based fluid and mechanical engineering simulations.

Research Intern, *Indian Space Research Organisation* **Trivandrum, India** *05/2018 – 05/2019*

- Developed a malware detection framework leveraging API call logs from 200+ malware and 100+ benign samples, proposing an API Call Transition Matrix (API-CTM) for feature extraction and achieving 96% detection accuracy. [Relevant publication]
- Built a controlled botnet testbed and generated network traffic datasets for cybersecurity research, analyzing HTTP/TCP headers to develop an N-gram-based detection technique achieving 99.5% accuracy in botnet traffic classification,

Organizational Experience

AWS Student Ambassador, *School of Management, Univ. Texas Dallas* *08/2024 – present*

- Conducted workshops and lectures on cloud infrastructure fundamentals, introducing core cloud services and operations, and organizing workshops, quizzes and peer competitions to enhance practical understanding of AWS among students.
- Created tutorials and guided hands-on projects for data visualization using Power BI, Tableau, and SQL, enabling students to build real-world analytical dashboards and improve data interpretation skills.
- Developed interactive educational websites for children focused on science and history, incorporating engaging visuals, quizzes, and activities to enhance foundational learning and curiosity.

Events Organizer, *Media & Streaming Systems, Satish Dhawan Space Center* *08/2020 – 12/23*

- Organized and supported events at Satish Dhawan Space Center and affiliated schools and communities, providing audio-video setup and live streaming services for scientific, academic, and cultural programs.
- Coordinated sports and community initiatives, including scheduling, venue planning, and approvals, while conducting environmental safety and cybersecurity seminars and leading community clean-up drives.

Web & Events Coordinator, *Conscientia & Dhanak, IIST* *10/2017 – 10/2018*

- Led web development and publicity teams for college festivals, developing event websites, organizing online events and coding hackathons, and managing registrations, resulting in a 45% increase in participant engagement.
- Contributed to festival planning, including event execution support, campus decoration, and participant hospitality, ensuring smooth coordination and positive on-ground experience for attendees.